



Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
Institut Teknologi Sepuluh Nopember

Laporan Akhir

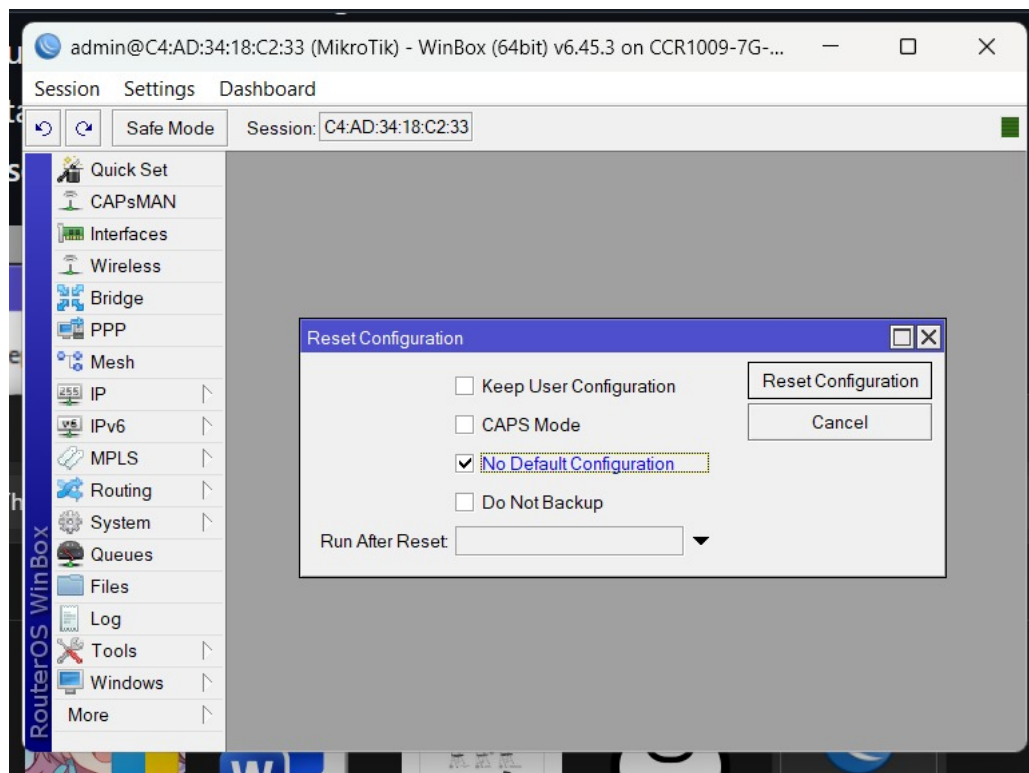
Praktikum Jaringan Komputer

Firewall dan NAT

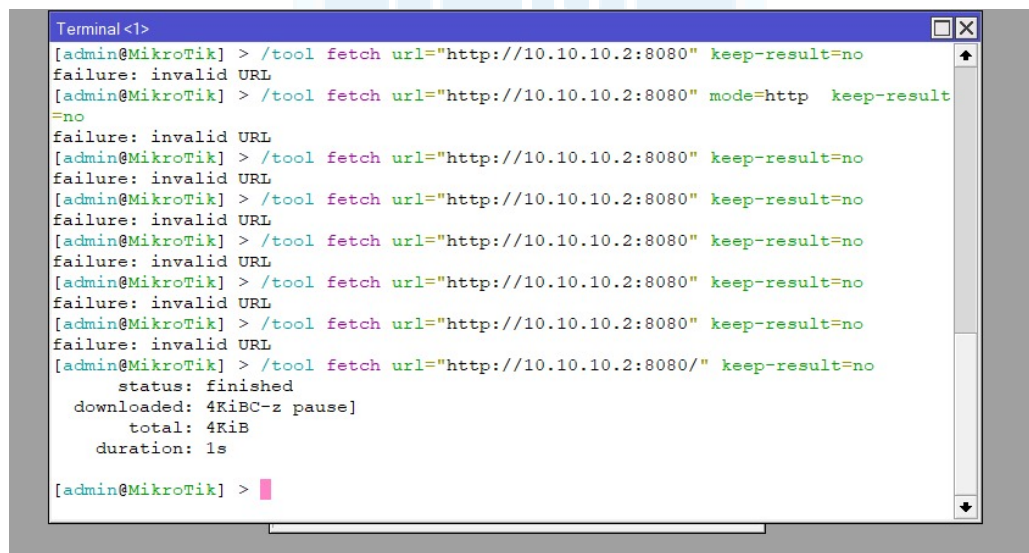
Riggy Fahmi Abyan - 5024241045

1 Juni 2026

1 Langkah-Langkah Percobaan



Gambar 1: reset configuration



Gambar 2: fetch url

2 Analisis Hasil Percobaan

Dari hasil percobaan , bisa dilihat bahwa firewall bisa memblokir ataupun melanjutkan pengiriman paket data dari ip tertentu.

```
[admin@MikroTik] > ping 10.10.10.2
SEQ HOST                                SIZE TTL TIME  STATUS
0 10.10.10.2                            56  64 0ms
1 10.10.10.2                            56  64 0ms
2 10.10.10.2                            56  64 0ms
3 10.10.10.2                            56  64 0ms
sent=4 received=4 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=0ms

[admin@MikroTik] > ping 192.168.10.2
SEQ HOST                                SIZE TTL TIME  STATUS
0 192.168.10.2                          56  127 timeout
1 192.168.10.2                          56  127 timeout
2 192.168.10.2                          56  127 timeout
3 192.168.10.2                          56  127 timeout
sent=4 received=0 packet-loss=100%

[admin@MikroTik] >
```

Gambar 3: ping firewall

```
Terminal <1>
[admin@MikroTik] > ping 192.168.10.2
SEQ HOST                                SIZE TTL TIME  STATUS
0 192.168.10.2                            56  127 0ms
1 192.168.10.2                            56  127 0ms
sent=2 received=2 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=0ms

[admin@MikroTik] > ping 10.10.10.2
SEQ HOST                                SIZE TTL TIME  STATUS
0 10.10.10.2                              56  127 timeout
1 10.10.10.2                              56  127 timeout
2 10.10.10.2                              56  127 timeout
3 10.10.10.2                              56  127 timeout
sent=4 received=0 packet-loss=100%

[admin@MikroTik] > ping 192.168.10.2
SEQ HOST                                SIZE TTL TIME  STATUS
0 192.168.10.2                            56  127 0ms
1 192.168.10.2                            56  127 0ms
2 192.168.10.2                            56  127 0ms
3 192.168.10.2                            56  127 0ms
sent=4 received=4 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=0ms

[admin@MikroTik] >
```

Gambar 4: masih bagian firewall

3 Hasil Tugas Modul

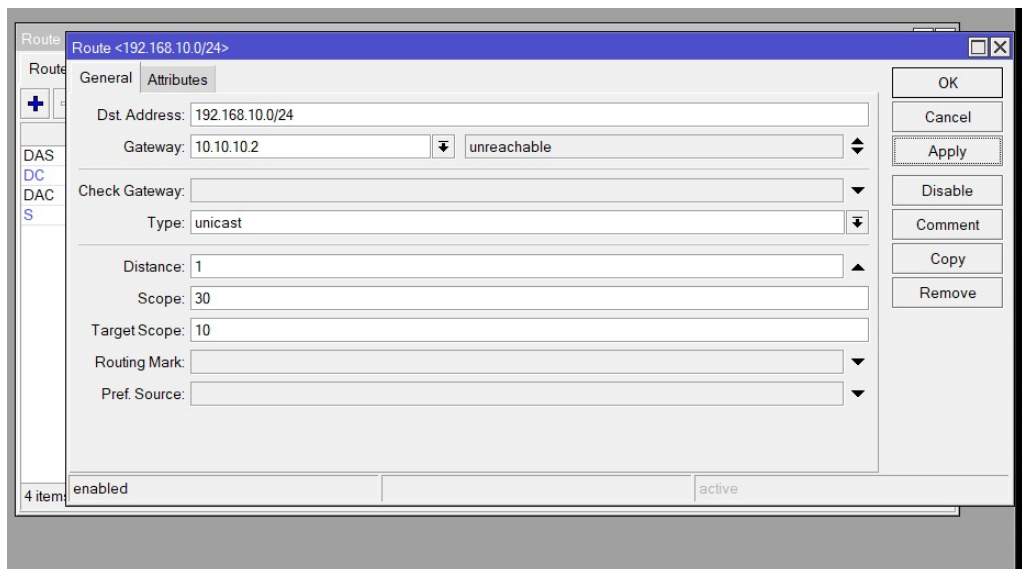
-

4 Kesimpulan

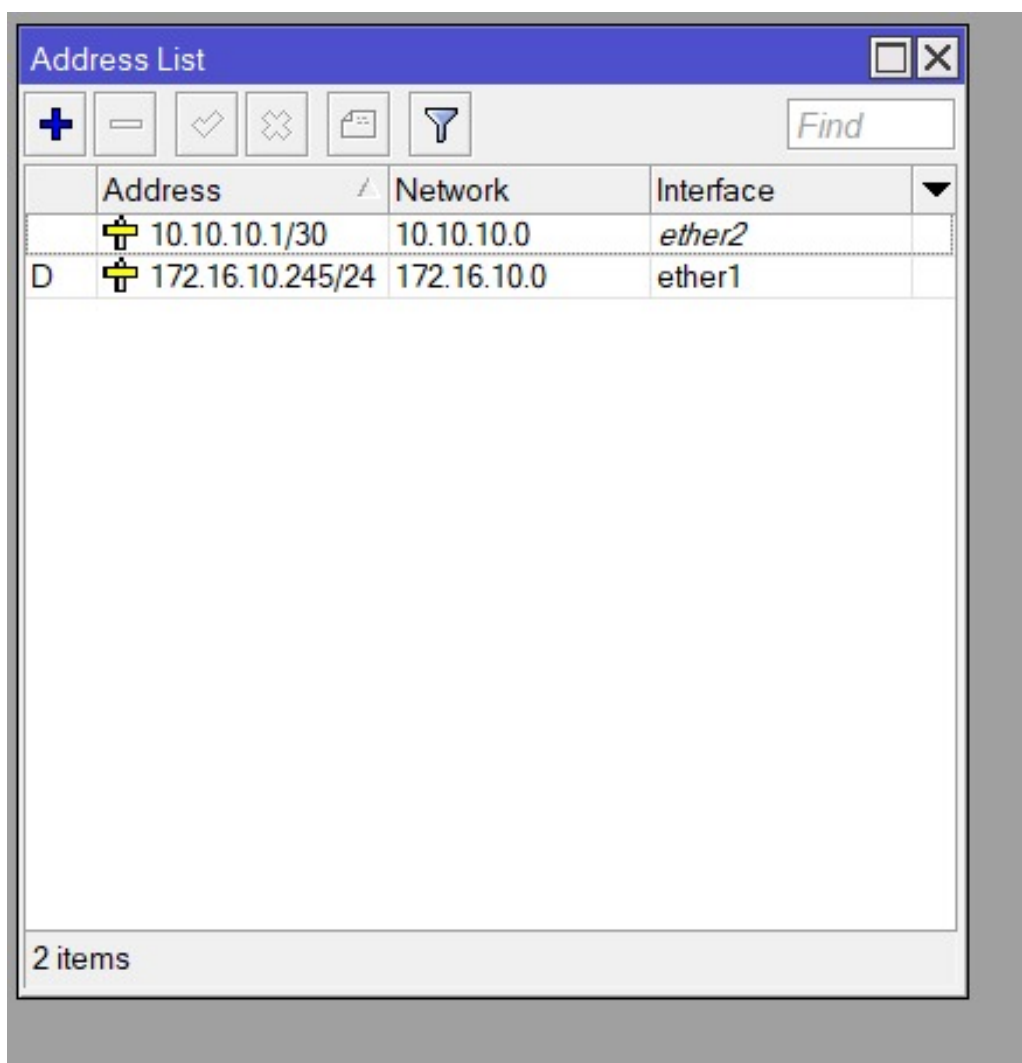
Dari paraktikum bisa dipelajari bahwa firewall bisa digunakan untuk menblokir ip-ipn yang tidak diinginkan dalam ajringan

5 Lampiran

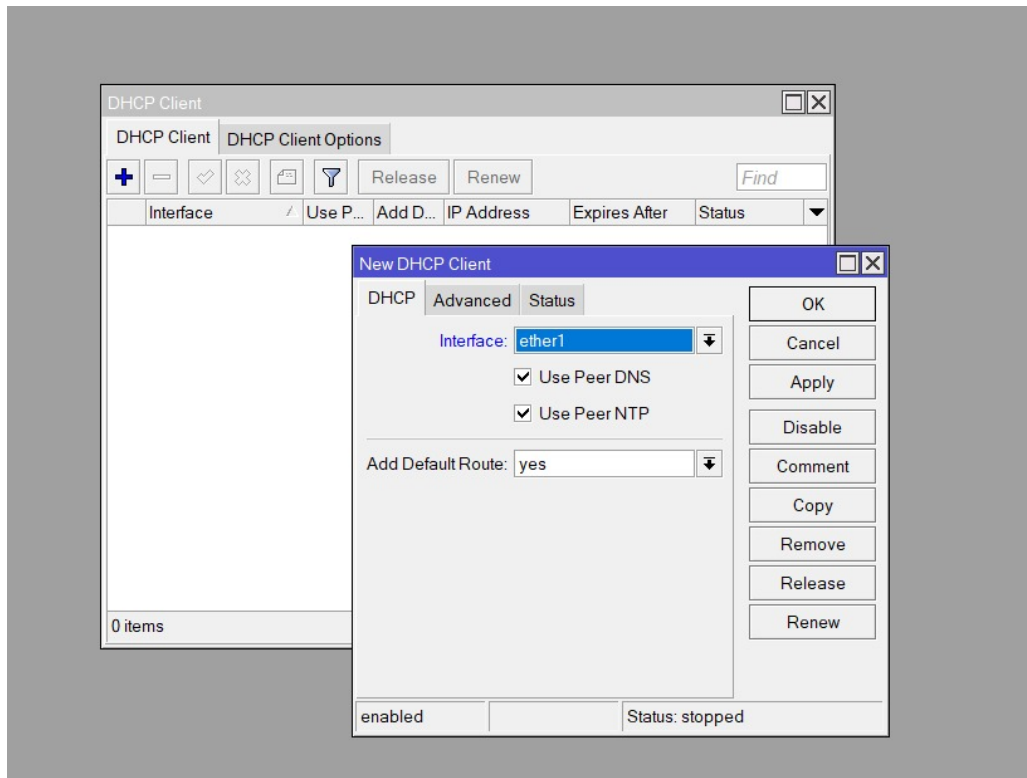
5.1 Dokumentasi saat praktikum



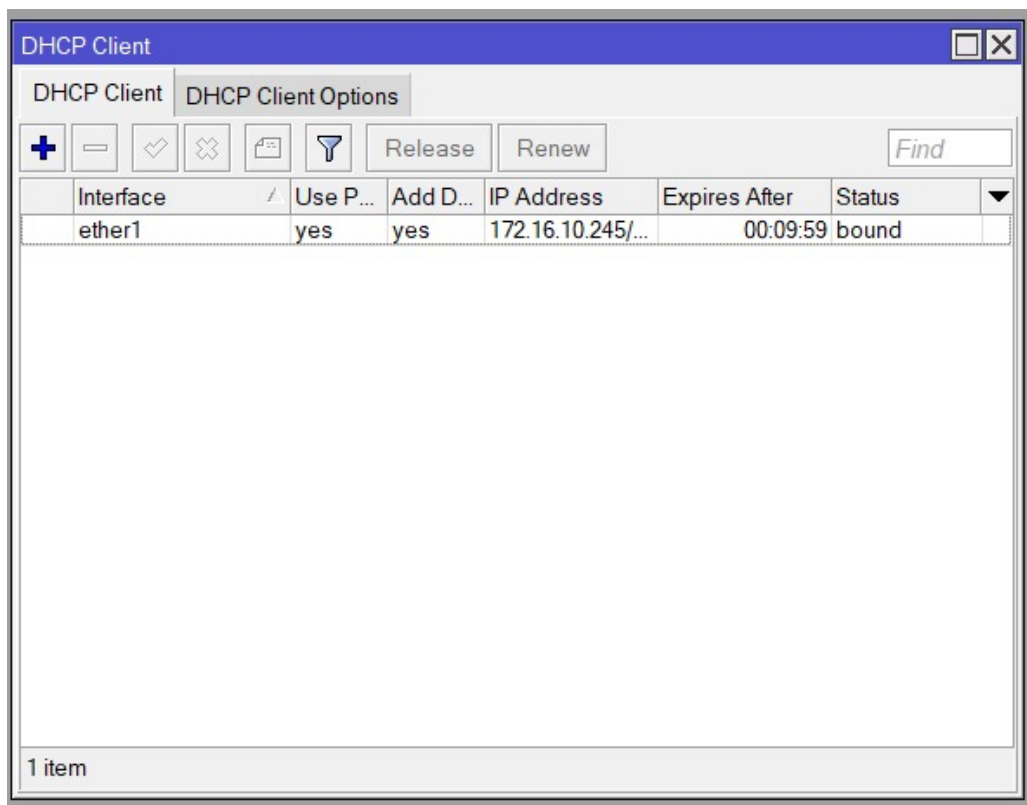
Gambar 5: static routing



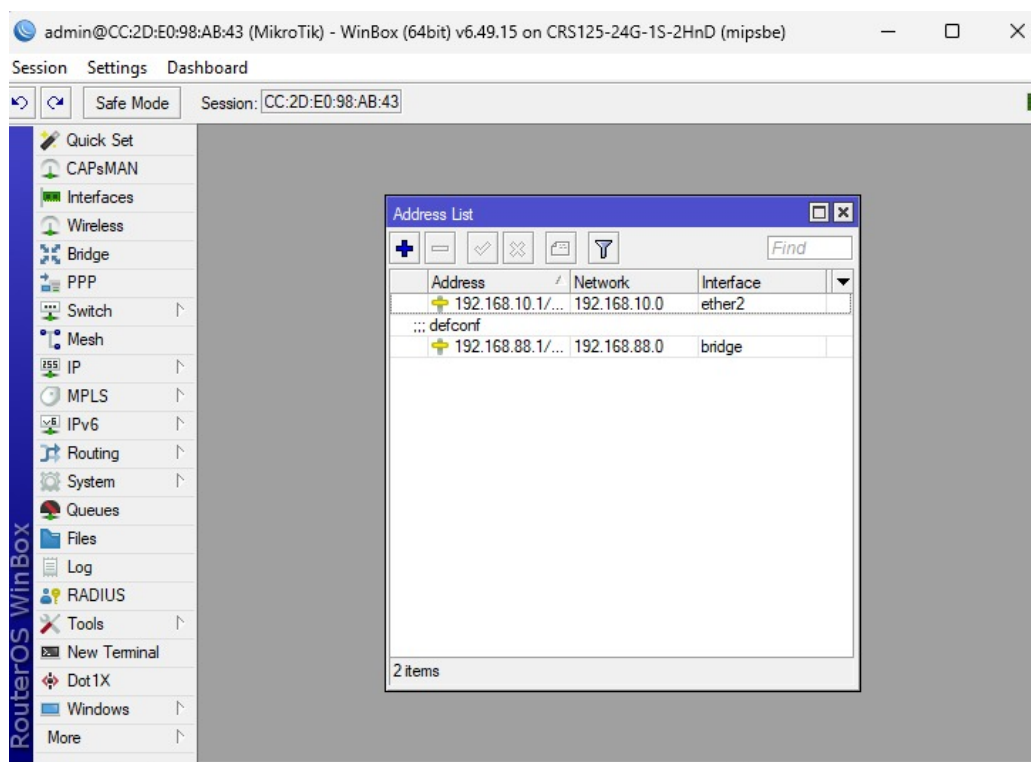
Gambar 6: IP



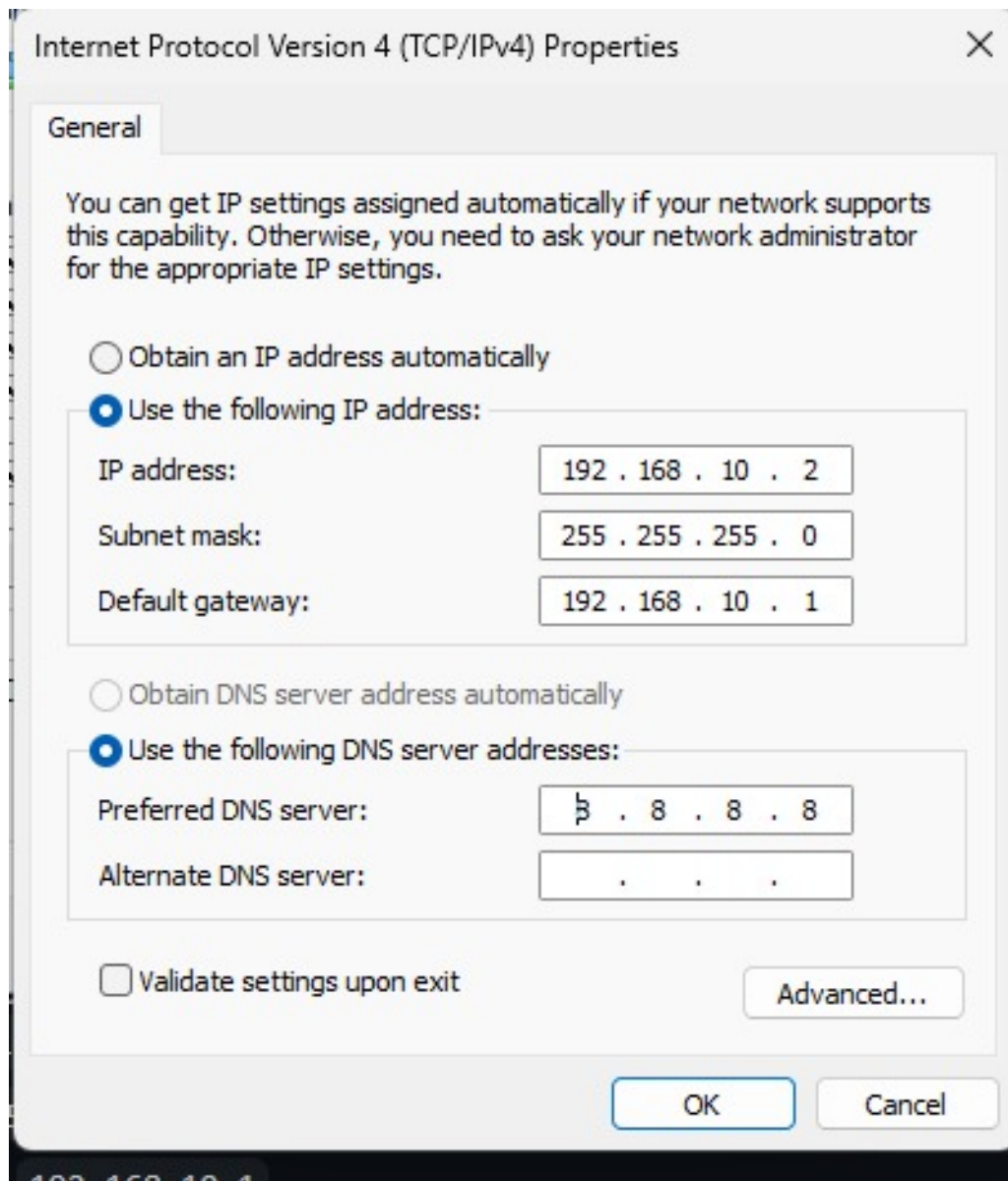
Gambar 7: set dhcp



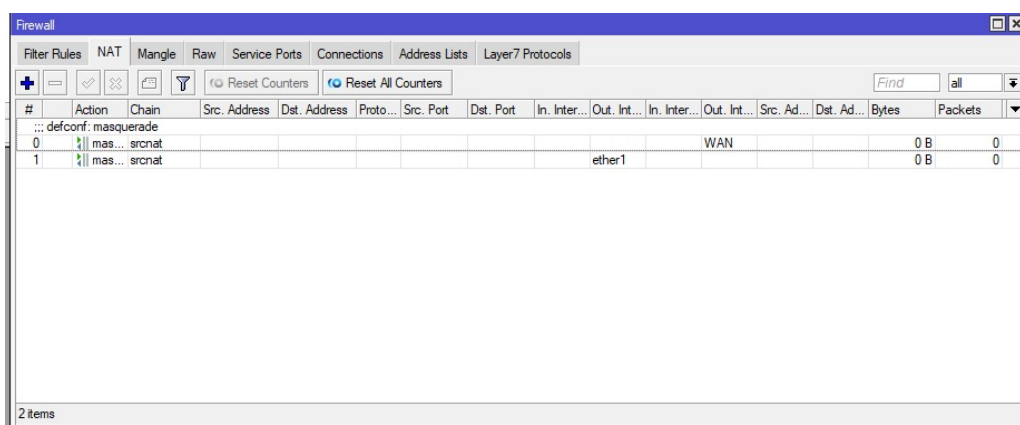
Gambar 8: dhcp client



Gambar 9: ip address



Gambar 10: set ip laptop



Gambar 11: konfigurasi NAT

Route List				
Routes				
Nextthops				
Rules				
VRF				
+ - ✓ ✗ 📄 🔍 Find all ↓				
	Dst. Address	Gateway	Distance	Routing Mark
S	0.0.0.0/0	10.10.10.1 unreachable	1	
DC	10.10.10.0/30	ether1 unreachable	255	
DAC	192.168.10.0/...	bridge reachable	0	
DAC	192.168.88.0/...	bridge reachable	0	

Gambar 12: routing

```

Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Users\Riggy>ping 172.16.10.245

Pinging 172.16.10.245 with 32 bytes of data:
Reply from 172.16.10.245: bytes=32 time<1ms TTL=63
Reply from 172.16.10.245: bytes=32 time<1ms TTL=63
Reply from 172.16.10.245: bytes=32 time<1ms TTL=63
Reply from 172.16.10.245: bytes=32 time<1ms TTL=63

Ping statistics for 172.16.10.245:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Riggy>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Riggy>

```

Gambar 13: ping

New Firewall Rule

General Advanced Extra Action Statistics

Chain:

Src. Address: ☐

Dst. Address:

Protocol: ☐

Src. Port:

Dst. Port:

Any. Port:

In. Interface:

Out. Interface:

In. Interface List:

Out. Interface List:

Packet Mark:

Connection Mark:

Routing Mark:

Routing Table:

Connection Type:

Connection State:

Connection NAT State:

enabled

OK

Cancel

Apply

Disable

Comment

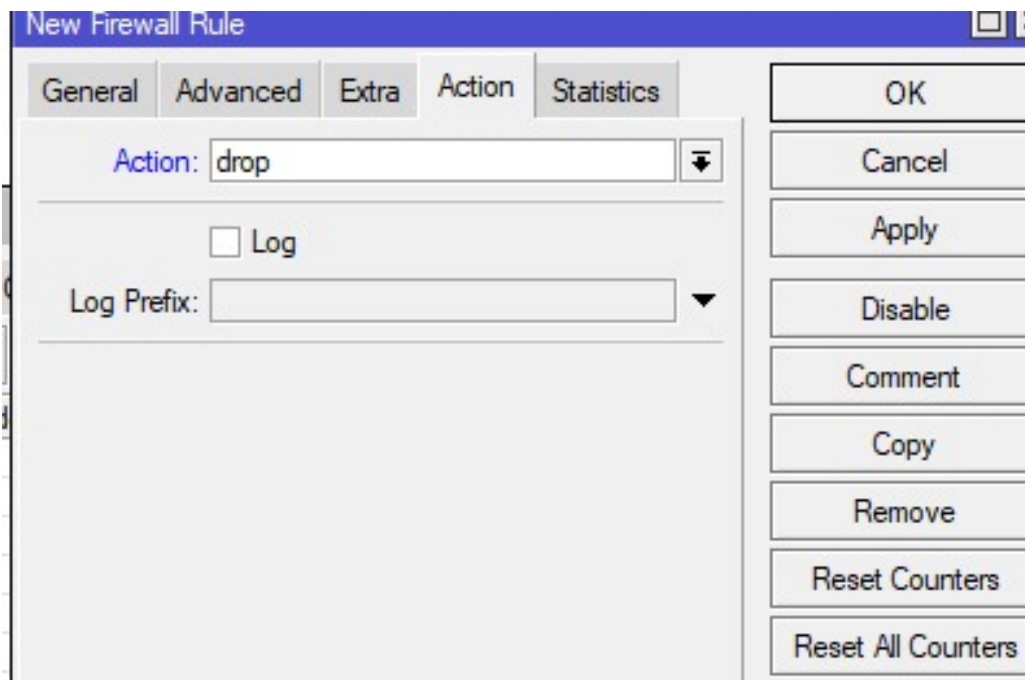
Copy

Remove

Reset Counters

Reset All Counters

Gambar 14: membuat aturan firewall



Gambar 15: action

Filter Rules															
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Gambar 16: memastikan aturan sudah masuk daftar

New Firewall Rule

General Advanced Extra Action Statistics

Chain:

Src. Address: ☐

Dst. Address:

Protocol: ☐

Src. Port:

Dst. Port:

Any. Port:

In. Interface:

Out. Interface:

In. Interface List:

Out. Interface List:

Packet Mark:

Connection Mark:

Routing Mark:

Routing Table:

Connection Type:

Connection State:

Connection NAT State:

enabled

OK

Cancel

Apply

Disable

Comment

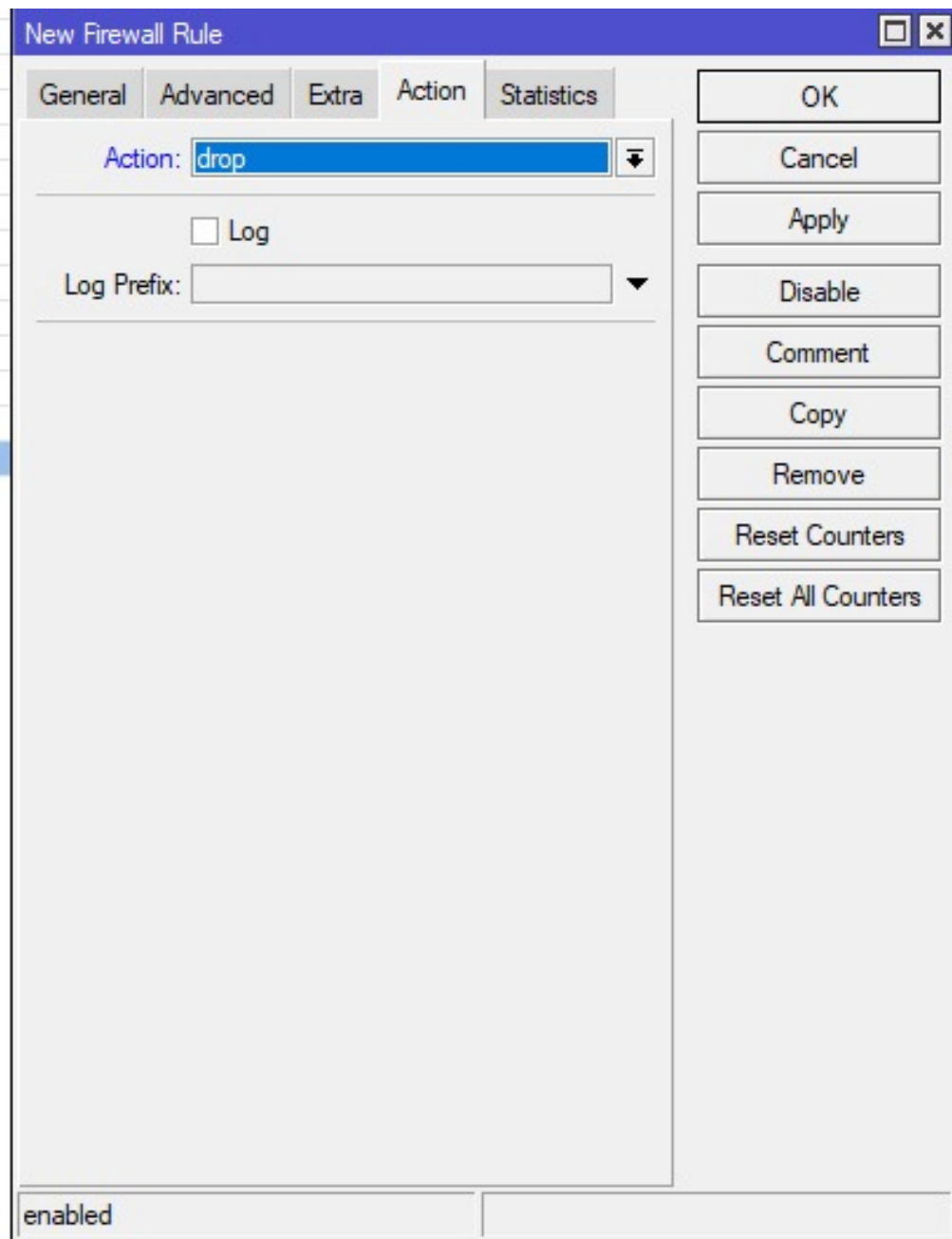
Copy

Remove

Reset Counters

Reset All Counters

Gambar 17: membuat aturan firewall



Gambar 18: action drop

New NAT Rule

General Advanced Extra Action ...

Chain: dstnat

Src. Address:

Dst. Address: 10.10.10.2

Protocol: 6 (tcp)

Src. Port:

Dst. Port: 8080

Any. Port:

In. Interface:

Out. Interface:

In. Interface List:

Out. Interface List:

Packet Mark:

Connection Mark:

Routing Mark:

Routing Table:

Connection Type:

enabled

OK

Cancel

Apply

Disable

Comment

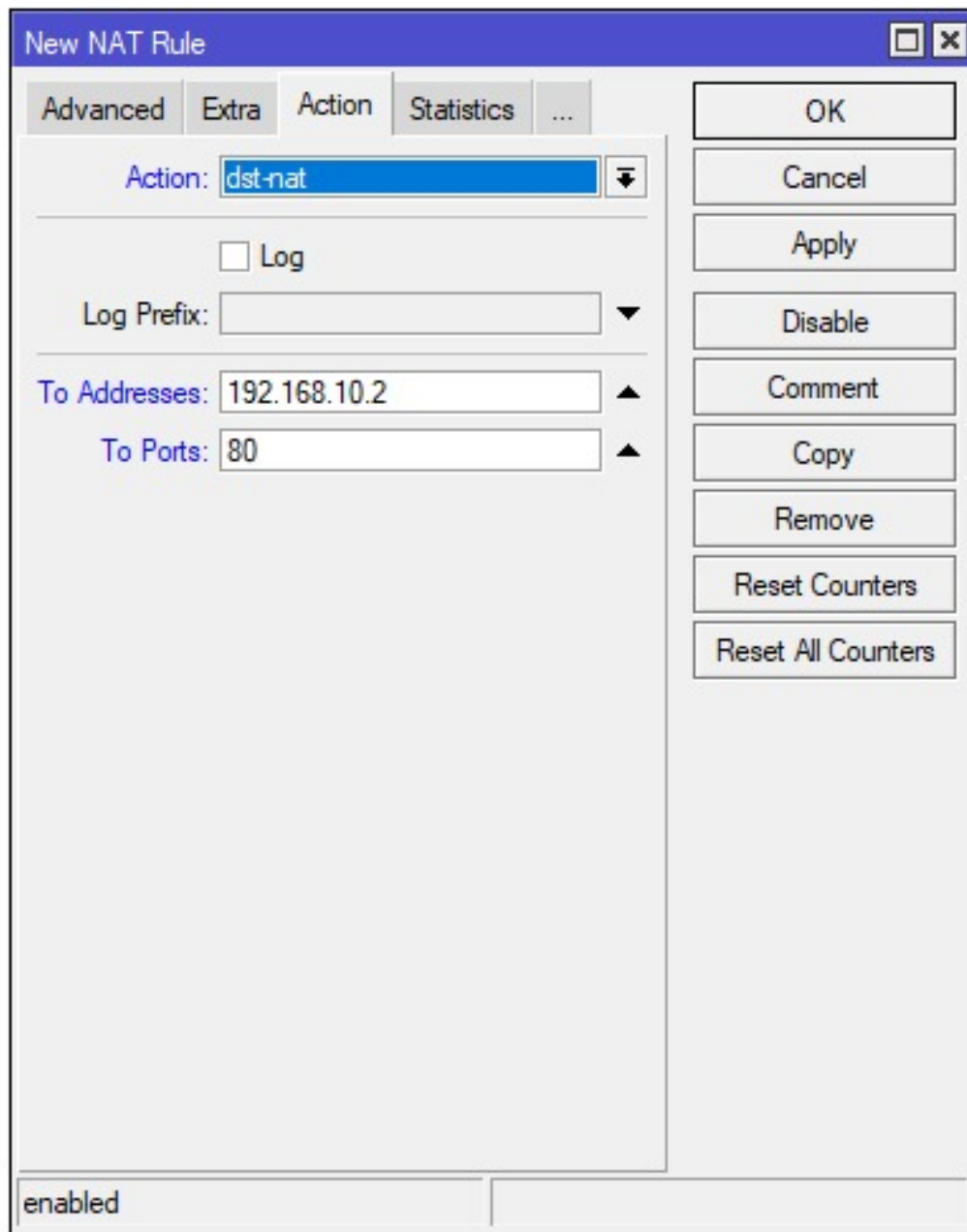
Copy

Remove

Reset Counters

Reset All Counters

Gambar 19: konfigurasi port forwarding



Gambar 20: action

```

Wireless LAN adapter Wi-Fi:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : its.ac.id

C:\Users\Riggy>python -m http.server 80
Serving HTTP on :: port 80 (http://[::]:80/) ...

```

Gambar 21: menyalakan server

```

Wireless LAN adapter Wi-Fi:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : its.ac.id

C:\Users\Riggy>python -m http.server 80
Serving HTTP on :: port 80 (http://[::]:80/) ...

```

Gambar 22: firewall filter connection

```
Wireless LAN adapter Wi-Fi:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : its.ac.id

C:\Users\Riggy>python -m http.server 80
Serving HTTP on :: port 80 (http://[::]:80/) ...
|
```

Gambar 23: ping

```
Wireless LAN adapter Wi-Fi:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : its.ac.id

C:\Users\Riggy>python -m http.server 80
Serving HTTP on :: port 80 (http://[::]:80/) ...
|
```

Gambar 24: Deskripsi Gambar

	Src. Address	Dst. Address	Proto...	Connecti...	TimeOut	TCP State	Orig./repl. Rate	Orig./repl. bytes	
Cs	192.168.10.250410	8.8.8.8:53	17 (u...		00:00:00	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.250521	8.8.8.8:53	17 (u...		00:00:02	0 bps/0 bps	67 B/0 B		
Cs	192.168.10.250544	8.8.8.8:53	17 (u...		00:00:02	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.250721	8.8.8.8:53	17 (u...		00:00:02	0 bps/0 bps	66 B/0 B		
Cs	192.168.10.250777	8.8.8.8:53	17 (u...		00:00:08	0 bps/0 bps	71 B/0 B		
Cs	192.168.10.250841	8.8.8.8:53	17 (u...		00:00:09	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.250843	8.8.8.8:53	17 (u...		00:00:05	0 bps/0 bps	67 B/0 B		
Cs	192.168.10.250844	8.8.8.8:53	17 (u...		00:00:00	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.250970	8.8.8.8:53	17 (u...		00:00:00	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.251045	8.8.8.8:53	17 (u...		00:00:02	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.251210	8.8.8.8:53	17 (u...		00:00:02	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.251304	8.8.8.8:53	17 (u...		00:00:01	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.251405	8.8.8.8:53	17 (u...		00:00:06	0 bps/0 bps	58 B/0 B		
Cs	192.168.10.251414	8.8.8.8:53	17 (u...		00:00:01	0 bps/0 bps	310 B/0 B		
Cs	192.168.10.251514	8.8.8.8:53	17 (u...		00:00:02	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.251614	8.8.8.8:53	17 (u...		00:00:01	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.251683	8.8.8.8:53	17 (u...		00:00:01	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.251837	8.8.8.8:53	17 (u...		00:00:01	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.251841	8.8.8.8:53	17 (u...		00:00:02	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.251956	8.8.8.8:53	17 (u...		00:00:00	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.251982	8.8.8.8:53	17 (u...		00:00:02	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.252008	8.8.8.8:53	17 (u...		00:00:03	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.252049	8.8.8.8:53	17 (u...		00:00:02	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.252268	8.8.8.8:53	17 (u...		00:00:09	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.252322	8.8.8.8:53	17 (u...		00:00:03	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.252408	8.8.8.8:53	17 (u...		00:00:01	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.252432	8.8.8.8:53	17 (u...		00:00:09	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.252583	8.8.8.8:53	17 (u...		00:00:00	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.252610	8.8.8.8:53	17 (u...		00:00:07	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.252657	8.8.8.8:53	17 (u...		00:00:09	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.252704	8.8.8.8:53	17 (u...		00:00:09	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.252999	8.8.8.8:53	17 (u...		00:00:02	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.253038	8.8.8.8:53	17 (u...		00:00:09	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.253057	8.8.8.8:53	17 (u...		00:00:09	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.253067	8.8.8.8:53	17 (u...		00:00:08	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.253104	8.8.8.8:53	17 (u...		00:00:02	0 bps/0 bps	62 B/0 B		
Cs	192.168.10.253419	8.8.8.8:53	17 (u...		00:00:08	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.253494	8.8.8.8:53	17 (u...		00:00:02	0 bps/0 bps	75 B/0 B		
Cs	192.168.10.253504	8.8.8.8:53	17 (u...		00:00:03	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.253609	8.8.8.8:53	17 (u...		00:00:09	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.253967	8.8.8.8:53	17 (u...		00:00:00	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.254010	8.8.8.8:53	17 (u...		00:00:02	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.254136	8.8.8.8:53	17 (u...		00:00:07	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.254195	8.8.8.8:53	17 (u...		00:00:06	0 bps/0 bps	66 B/0 B		
Cs	192.168.10.254237	8.8.8.8:53	17 (u...		00:00:02	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.254341	8.8.8.8:53	17 (u...		00:00:09	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.254519	8.8.8.8:53	17 (u...		00:00:09	0 bps/0 bps	60 B/0 B		
Cs	192.168.10.254590	8.8.8.8:53	17 (u...		00:00:09	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.254690	8.8.8.8:53	17 (u...		00:00:09	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.254784	8.8.8.8:53	17 (u...		00:00:02	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.255026	8.8.8.8:53	17 (u...		00:00:01	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.255042	8.8.8.8:53	17 (u...		00:00:02	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.255084	8.8.8.8:53	17 (u...		00:00:06	0 bps/0 bps	67 B/0 B		
Cs	192.168.10.255207	8.8.8.8:53	17 (u...		00:00:02	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.255237	8.8.8.8:53	17 (u...		00:00:07	0 bps/0 bps	71 B/0 B		
Cs	192.168.10.255333	8.8.8.8:53	17 (u...		00:00:08	0 bps/0 bps	56 B/0 B		
Cs	192.168.10.255397	8.8.8.8:53	17 (u...		00:00:02	0 bps/0 bps	56 B/0 B		

Gambar 25: Deskripsi Gambar


```
Object "adwadwafwafawfawf" is unknown, try "ip help".
root@kvm:~#
root@kvm:~#
root@kvm:~#
root@kvm:~# ping 192.168.20.1 -c 4
PING 192.168.20.1 (192.168.20.1) 56(84) bytes of data.
64 bytes from 192.168.20.1: icmp_seq=1 ttl=255 time=0.578 ms
64 bytes from 192.168.20.1: icmp_seq=2 ttl=255 time=0.294 ms
64 bytes from 192.168.20.1: icmp_seq=3 ttl=255 time=0.247 ms
64 bytes from 192.168.20.1: icmp_seq=4 ttl=255 time=0.365 ms

--- 192.168.20.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3069ms
rtt min/avg/max/mdev = 0.247/0.371/0.578/0.126 ms
root@kvm:~#
```

Gambar 26: Deskripsi Gambar



Gambar 27: Deskripsi Gambar